

# Expressiveness, Separation and Complexity Analyses on Multiparty Session Types

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*Multiparty session types* (MPST) are a type discipline for concurrent and distributed systems that ensures safety, deadlock freedom, and liveness of communicating components. This paper surveys recent advances in the *expressiveness*, *separation*, and *complexity* of MPST, with a particular focus on recent results in [62, 73, 35, 64, 51].

## 1 Introduction

*Multiparty session types* (MPST) [33, 32] are a type discipline for designing and verifying communication protocols involving multiple participants in concurrent and distributed systems. This paper discusses the *expressiveness* of the MPST framework.

In the EXPRESS community, the term “expressiveness” usually means “*encodability*” between process calculi [27]. An *encodability* result establishes an encoding from a *source calculus* to a *target calculus* that satisfies a given set of criteria, showing that the target can emulate the behaviour of the source and is therefore at least as expressive. Conversely, a *separation* result proves that *no such encoding exists*, demonstrating that certain behaviours of the source cannot be enumerated by the target. Encodability and separation results establish expressiveness hierarchies among calculi, provided they are based on the same correctness criteria.

Peters and Yoshida [62] have proved that a synchronous *mixed choice multiparty session calculus*, typed using the bottom-up strategy (see the next section), is strictly more expressive than multiparty session calculi with only *separate choice*. Here, *mixed choice* allows one choice to contain both input and output actions, but *separate choice* contains either input or output actions in the same summand. The same work also established that the *binary* mixed session calculus [6] is strictly less expressive, separating the expressive power of binary and multiparty session types.

More recently, Masters and Yoshida [51] have proved that the subtyping (refinement) relation of [62] is *precise*: the subtyping rules accept *exactly* the safe subtype relation while rejecting every unsafe one. Consequently, the typing system of [62] achieves the maximum expressiveness among synchronous mixed choice session calculi. But what does this notion of “maximum expressiveness” actually mean?

The remainder of this paper discusses expressiveness from a different perspective: the expressive power of *typing systems* with subtyping relations. Here, expressiveness refers to the *typability* of processes under two different typing systems with the same syntax and operational semantics. If the two typing systems yield exactly the same set of typable processes, they are said to have the same expressive power.

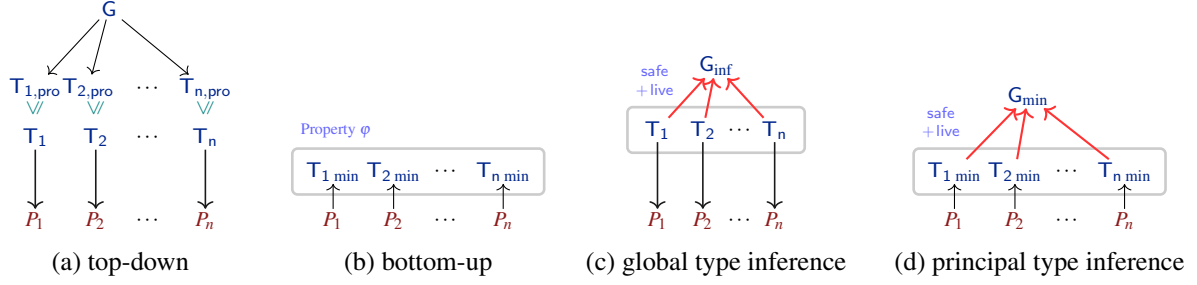


Figure 1: Multiparty session type frameworks from [64, Fig.1]

## 2 Expressiveness as Typability

This section discusses the expressiveness of two typing systems, namely the top-down and bottom-up typing systems.

### 2.1 Top-Down Strategy vs Bottom-Up Strategy

**Top-Down Strategy.** The original MPST theory [32, 33] takes the *top-down* strategy (Figure 1a). Instead of developing multiple distributed programs independently and hoping that their communications match correctly, a protocol designer first specifies the overall communication protocol—who sends what to whom, and with what type—as a *global type*  $G$ . An *endpoint projection* (EPP) algorithm then generates a set of local types  $\{T_{i,pro}\}_{i \in I}$ , each prescribing how an individual program  $P_i$  may communicate with the others.

Type checking guarantees that the resulting processes  $\{P_i\}_{i \in I}$  satisfy *communication safety* (no unexpected type or label mismatches), *session fidelity* (each process conforms to the specified global protocol), and *deadlock freedom*. In the top-down approach, typability also guarantees the stronger property of *liveness*, which means that every request eventually receives a response and that each participant can continue to make progress without stalling indefinitely.

The top-down strategy has been integrated into numerous programming languages and software frameworks through code generation, API generation and inference, event-driven programming, static type checking, protocol compliance, real-time systems, and runtime monitoring. Representative implementations include Rust [13, 46, 34, 74], Java [36, 37, 45, 4], Scala [65, 75, 12, 2, 19, 41], Go [9], TypeScript [53, 23], PureScript [44], OCaml [77, 39], MPI-C [58, 50], F $\star$  [78], F $\sharp$  [54], Python [57, 56, 14], Erlang [20, 55, 15, 3, 22], C [60, 61], C $\sharp$  [43], and domain-specific languages [48, 10, 11, 28, 21]. A comprehensive survey is given in [76].

The correctness of the top-down framework has also been verified in several theorem-proving frameworks [7, 71, 72, 40, 16, 49, 8, 42].

**Bottom-Up Strategy.** The bottom-up strategy, proposed in [66], directly verifies that a collection of local types—either written by the programmer or inferred from a set of programs—satisfies a desired property  $\phi$  (Figure 1b). The motivation of [66] was to develop a general MPST framework that does not rely on prescribing a global type, thereby establishing *type soundness* without relying on global types or endpoint projection (EPP). The paper further argued that existing proofs of type soundness based on endpoint projection with the *mergeability* operator are flawed. Since mergeability enlarges the class of

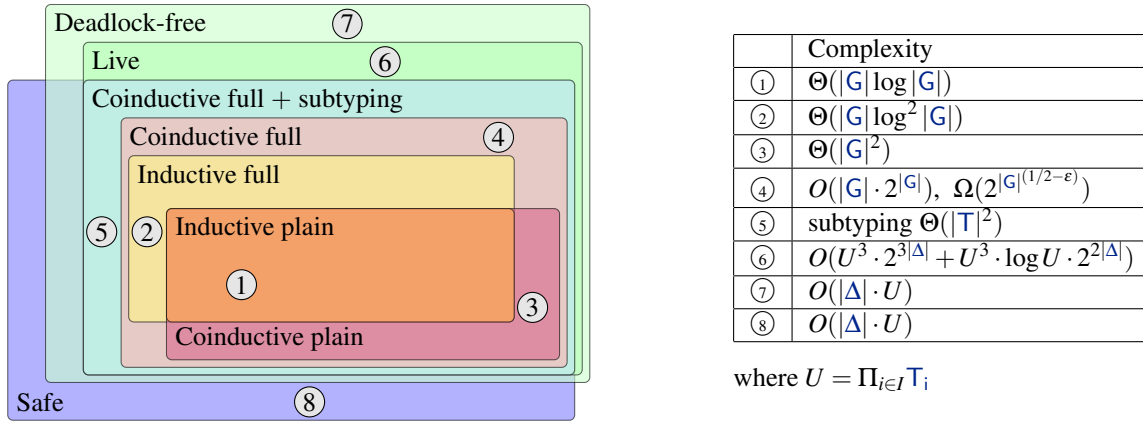


Figure 2: Projections, typing context properties and complexity from [73, Fig 4 and Tables 1,2,3]

typable endpoint programs and greatly simplifies implementations, this paper has given considerable attention.

Subsequent work [35] has resolved the above concern by introducing a general proof technique for type soundness (subject reduction) of the multiparty session  $\pi$ -calculus. The proof is based on an *association relation* between the behavioural semantics of a global type and that of its endpoint projections. Using this relation, the authors also established session fidelity, deadlock freedom, and liveness for endpoint projection with mergeability.

Nevertheless, the bottom-up strategy offers several distinctive advantages. First, it supports the verification of a broad range of behavioural properties. For example, it can verify communication safety independently of deadlock-freedom or liveness. Secondly, it yields the *maximum* typability with respect to a desired property  $\phi$  (see below for a formal definition). Thirdly, because local types closely resemble process calculi (e.g., CCS [52] and CSP [31]) or finite-state machines, the bottom-up approach can directly exploit mature off-the-shelf model checkers, such as mCRL2 [69] and SPIN [70], to verify behavioural properties.

The bottom-up approach has been integrated into Scala [1, 67], OCaml [38], Go [59, 47], and Rust [13, 46]. The bottom-up typing system has also been verified by Rocq [29, 17, 18, 30] and Why3 [26].

**Limitations of Top-Down Strategy.** While [35] solves some concerns in the literature, there is one shortcoming of the top-down strategy—*typability is limited* since it depends on the EPP algorithm being used. If  $G$  is not projectable by the chosen EPP algorithm, either  $G$  is not *implementable* (there exists no correct set of local types) or this particular EPP algorithm rejects some implementable  $G$ . If the EPP is too limited, many safe and live processes become untypable. As expected, computationally cheaper EPP algorithms tend to give fewer typable processes than more expensive ones [73]. See 2.

**Limitations of the Bottom-Up Strategy.** Since the bottom-up strategy directly verifies behavioural properties of local types, its main drawback is its *computational cost*. Verifying safety, deadlock-freedom and liveness of a collection of local types is PSPACE-complete in the synchronous setting [73] and *undecidable* in the asynchronous setting [5]. By contrast, the top-down strategy requires only endpoint projection and subtyping, both of which can be decided for most of the usecases, polynomial time in the size of the types [73].

**The First Summary.** The top-down and bottom-up strategies offer complementary trade-offs. The top-down strategy requires an upfront global protocol specification and endpoint projection, but enables computationally efficient verification and provides a human-readable protocol specification. Hence we can guarantee the *protocol conformance*. In contrast, the bottom-up strategy avoids the need for a global specification, but incurs significantly higher verification costs, which are PSPACE-complete in the synchronous setting and undecidable in the asynchronous setting.

## 2.2 Expressiveness as Typability

Recent work by Pischke and Yoshida [64] has established a surprising result concerning the expressive power (typability) of the top-down and bottom-up typing systems with respect to the *liveness* property under synchronous semantics.

Consider a participant  $\mathbf{p}_i :: P_i$ , where process  $P_i$  implements participant  $\mathbf{p}_i$  and has session type  $T_i$ . A *multiparty session* is a parallel composition of participants

$$M = \mathbf{p}_1 :: P_1 \mid \cdots \mid \mathbf{p}_n :: P_n$$

The bottom-up typing judgement is defined as:  $\vdash^{\text{bot}} M : \Delta$  where  $\Delta$  is a *safe typing context* mapping each participant  $\mathbf{p}_i$  to its type  $T_i$ . “Safety” on  $\Delta$  is the minimum requirement needed to guarantee type soundness of  $M$ .

Using this judgement, we define the set of typable live sessions  $\mathbb{P}_{\text{bot}}$  as:

$$\mathbb{P}_{\text{bot}} = \{ M \mid \vdash^{\text{bot}} M : \Delta \text{ such that } \Delta \text{ is live} \}$$

Whenever  $\Delta$  is live, the session  $M$  satisfies liveness [66, 35]. Furthermore,  $\mathbb{P}_{\text{bot}}$  is preserved under reduction: if  $M \in \mathbb{P}_{\text{bot}}$  and  $M \rightarrow M'$ , then  $M' \in \mathbb{P}_{\text{bot}}$ . Consequently,  $\mathbb{P}_{\text{bot}}$  provides a *complete characterisation of liveness*: a multiparty session is typable and live if and only if it belongs to  $\mathbb{P}_{\text{bot}}$ .

For the top-down system, each local type  $T_i$  is obtained by projecting a global type  $G$  onto participant  $\mathbf{p}_i$ , written  $G \upharpoonright_{\mathbf{p}_i} T_i$ . We therefore define:

$$\mathbb{P}_{\text{top}} = \{ M \mid \vdash^{\text{top}} M : \Delta \text{ such that } G \upharpoonright_{\mathbf{p}_i} T_i \quad \Delta = \{ \mathbf{p}_i : T_i \}_{i \in I} \text{ for some balanced } G \}$$

By construction, we have:  $\mathbb{P}_{\text{top}} \subseteq \mathbb{P}_{\text{bot}}$ .

Depending on the EPP algorithm, the inclusion can be strict. That is, a session may belong to  $\mathbb{P}_{\text{bot}}$  but not  $\mathbb{P}_{\text{top}}$ , because the chosen EPP algorithm fails to project a global type that can implement safe and live processes.

The main result of [64] is that this gap disappears under three conditions:

$$\mathbb{P}_{\text{top}} = \mathbb{P}_{\text{bot}}$$

This overturns the common belief that the top-down strategy is inherently incomplete. More precisely, the top-down system is complete with respect to liveness provided that (i) typing includes the subsumption rule induced by *precise synchronous subtyping*  $\leq$  [24]; (ii) global types are *balanced*, meaning every participant occurring in a subterm is unavoidable; and (iii) endpoint projection is performed using the most expressive (and computationally most expensive) projection algorithm proposed so far, namely the *coinductive full-merging projection* [73].

The proof of this result relies on a *principal global type inference algorithm*, developed in [64], which synthesises a balanced global type from any safe and live typing context (Figure 1c). The details of the synthesis algorithm and its implementation are found in [64].

### 3 Further Questions on Expressiveness

The equivalence result in [64] establishes that the top-down and bottom-up typing systems have the same expressive power (typability) only for the synchronous MPST calculus with *separated choice*, *??*, and *the liveness property*. This raises the question of whether similar equivalence results hold under other semantics, language features, subtyping relations, and behavioural properties.

A first future direction is *asynchronous semantics*. Pischke et al. [63] have recently extended the top-down framework with coinductive full-merging projection and precise asynchronous subtyping [25], establishing subject reduction and liveness for the asynchronous MPST-calculus by relying on the corresponding bottom-up results. An important open question is whether the top-down and bottom-up typing systems also coincide with respect to asynchronous liveness.

A second direction is to consider behavioural properties beyond liveness. The equivalence in [64] is established only for liveness. Whether an analogous result holds for deadlock-freedom remains open, i.e., whether the top-down typing system can completely characterise the deadlock-free processes accepted by the bottom-up framework (if we change the balanced condition).

A third direction is more expressive process calculi (in the sense of encodability). The current expressiveness result is limited to separated choice. Extending it to mixed-choice MPST appears substantially more difficult. In the asynchronous setting, Stutz proved that deciding whether a global type with mixed choice admits a deadlock-free implementation is undecidable [68]. The same argument might be true for synchronous semantics. Consequently, no decidable endpoint projection algorithm can completely characterise the deadlock-free behaviour of local types. It is therefore unlikely that any decidable projection algorithm can completely capture liveness for mixed-choice MPST even up to the precise mixed choice subtyping [51]. This suggests that practical top-down systems will necessarily rely on projection algorithms that have given up completeness for decidability and efficiency, as proposed in [3].

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